

OLLSCOIL NA hÉIREANN, CORCAIGH
THE NATIONAL UNIVERSITY OF IRELAND, CORK

COLÁISTE NA hOLLSCOILE, CORCAIGH
UNIVERSITY COLLEGE, CORK

Examination Session and Year	Autumn 2023
Module Code	AC2204
Module Title	INTRODUCTION TO MANAGEMENT ACCOUNTING
Paper Number	1
External Examiner	Professor Martin Quinn
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Instructions to Candidates	Answer three questions in total. Please attempt QUESTION ONE from Section A and TWO questions from Section B.
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided they are not programmable and can not store text (other than algebraic notation).

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SECTION A: COMPULSORY

QUESTION ONE

Split Ltd commenced operations on 1 January 2021. The managing director is not used to preparing accounting calculations and has asked for your help and expertise. The company manufactures and sells a single product for which the following information for the 2021 financial period is available.

Sales (in units)	5,240
Production (in units)	6,800
Selling price per unit	€34
Variable manufacturing costs (total)	€106,080
Fixed manufacturing costs (total)	€59,160
Variable marketing costs (total)	€19,340
Fixed marketing costs (total)	€13,600

Required:

- (A) Calculate the unit product cost, the value of closing stock and the operating profit using each of the following methods:

- (i) Marginal costing;
- (ii) Absorption costing.

[Please show the income statement for each method, as part of your answer]

(18 marks)

- (B) The managing director is confused by the figures calculated in Part A above and does not understand how it is possible to have two profit figures and two values for units in closing stock. Prepare a short note that explains why this is happening. Include in your answer calculations (profit reconciliation) which explain the difference in the operating profit figures resulting from the use of the two costing methods in Part A above.

(10 marks)

- (C) Identify and describe three differences between management accounting and financial accounting.

(12 marks)

(Total: 40 marks)

SECTION B
ANSWER ANY TWO QUESTIONS

QUESTION TWO

Morgan Ltd currently uses a traditional (absorption) costing system to determine the cost of each of its three products. Production overheads (totalling €252,000) are absorbed on the basis of machine hours, at a rate of €36 per machine hour. Direct labour costs €27 per labour hour. The following additional data is available for the most recent financial period.

	Total production volume (units)	Direct Labour hours per unit	Machine hours per unit	Total machine hours	Direct Material cost per unit
Product A	1,000	2 hours	2.0 hours	2,000	€25
Product B	2,000	3 hours	1.5 hours	3,000	€40
Product C	500	4 hours	4.0 hours	2,000	€10

Morgan Ltd is considering implementing an activity-based costing system. Management have estimated that production overheads can be divided into the cost pools shown below.

Cost Pool	% of total overhead	Cost Driver
Machining	30%	No. of machine hours
Production scheduling	25%	No. of set-ups
Quality control	30%	No. of units
Materials handling	15%	No. of purchase orders

Activity volumes for the financial period under review are detailed below.

	No. of purchase orders	No. of set-ups
Product A	10	5
Product B	5	20
Product C	10	10
Total	25	35

Required:

- (A) Calculate the total costs and the cost per unit for each product, using:
- (i) Conventional (absorption) costing, with production overheads allocated based on machine hours.
 - (ii) Activity-based costing, with production overheads allocated based on the cost drivers given in the question.
- [Please show all your workings clearly]

(21 marks)

- (B) Distinguish, with the aid of diagrams, between a variable cost, a fixed cost and a mixed cost. In your answer, give one example of each type of cost.

(9 marks)

Total: 30 marks)

QUESTION THREE

Management at Crisper Ltd have become increasingly concerned with the management accounting and financial planning practices in use at the organisation. You have been hired to help answer questions raised at a recent management meeting.

Details relating to the number of units manufactured and the total machine maintenance costs incurred for the past six years are as follows. Crisper Ltd wishes to use this data to estimate the maintenance costs for 2022.

	Units manufactured	Total machine maintenance costs
2016	34,000	€290,100
2017	45,000	€344,000
2018	37,900	€309,210
2019	40,000	€319,500
2020	31,500	€277,850
2021	42,000	€329,300

Required:

- (A) Using the High-Low method, estimate a cost function and use it to determine expected total machine maintenance costs for 2022, if 39,000 units are expected to be manufactured.

(10 marks)

- (B) List and explain two other methods that can be used to estimate the fixed and variable components of a mixed cost.

(8 marks)

- (C) Distinguish between each of the following terms widely used in management accounting:

- (i) Sunk costs and opportunity costs.
- (ii) Prime costs and conversion costs.
- (iii) Planning and control.

(12 marks)

(Total: 30 marks)

QUESTION FOUR

Consider each of the following statements. For each statement, state whether you agree or disagree, explaining your answer in each case.

- (A) Oak Ltd produces a single product. If total sales are 30,000 units, totalling €165,000; total variable expenses are €63,000 and total fixed costs are €56,100, then break-even point is 16,500 units, the margin of safety is 45% and operating leverage is 4.
- (B) Both the Operating Leverage and the Margin of Safety, at a given level of sales in a capital intensive company, will tend to be lower than in a labour intensive company, with the same total revenues and total expenses.
- (C) As production increases within the relevant range, fixed cost per unit decreases.
- (D) The contribution margin ratio measures the amount by which sales revenue can fall before losses begin to be incurred.
- (E) Alex makes two types of cakes. Based on the budgeted production and sales information shown in the table below for the next financial period, and assuming total fixed costs of €5,760, the breakeven point is 480 cherry cakes and 320 coconut cakes.

	Cherry cake	Coconut cake
Budgeted cake sales (units)	18,000	12,000
Budgeted sales price per cake	€20.00	€15.00
Variable costs per cake	€12.00	€9.00

(5 * 6 marks each)
(Total: 30 marks)

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